

Multiple choice Question with one option correct

Q.01 to Q.10

Multiple choice Question with one or more than one may be correct

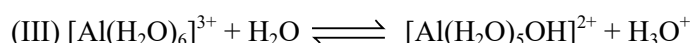
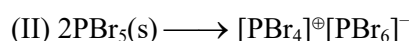
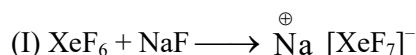
Q.11 to Q.30

Numerical Based Question (no negative marking)

Q.31 to Q.50

**DPP Syllabus : Metallurgy, p-Block, d & f-Block, Qualitative analysis**

1. Consider the following transformations :-



Possible transformations are :

- (A) I, II, III                      (B) I, III                      (C) I, II                      (D) II, III

2. A compound (P) on hydrolysis gives gas (Q) which on oxidation by bleaching powder gives (R). Gas (R) on reaction with magnesium gives compound (S) which on hydrolysis again gives gas (Q). Compound (P) can not be :-

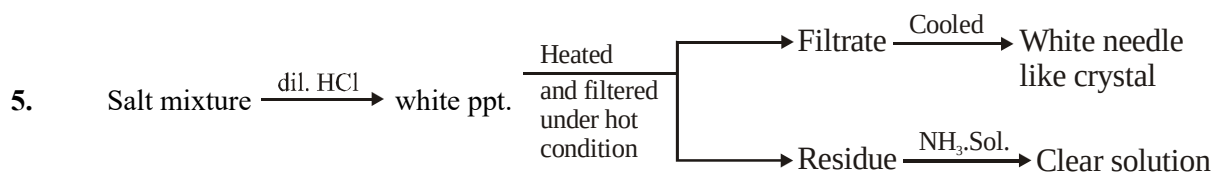
- (A) Calcium imide              (B) Lithium nitride              (C) Sodamide              (D) Calcium nitrate

3. When  $\text{F}_2$  is passed into a solution of mineral acid X, a greenish yellow gas Y is formed. Which on treating with slaked lime forms "Z". When Red litmus is kept in contact with Z, it changes into :-

- (A) Blue colour              (B) No change in colour      (C) White in colour              (D) None of these

4. The product of hydrolysis of  $\text{SF}_4$  and  $\text{TeF}_6$  are \_\_\_\_\_ and \_\_\_\_\_ respectively.

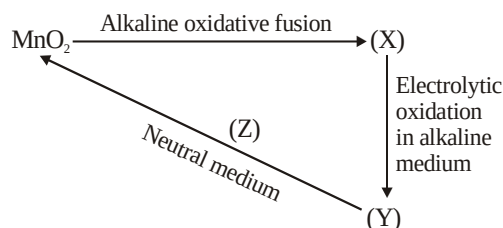
- (A)  $\text{H}_2\text{SO}_3$  and  $\text{H}_2\text{TeO}_4$                       (B)  $\text{H}_2\text{SO}_4$  and  $\text{H}_2\text{TeO}_4$   
 (C)  $\text{H}_2\text{SO}_3$  and  $\text{Te}(\text{OH})_6$                       (D)  $\text{H}_2\text{SO}_4$  and  $\text{Te}(\text{OH})_6$



Salt is consisting of cations :

- (A)  $\text{Pb}^{2+}$  and  $\text{Hg}^{2+}$               (B)  $\text{Ag}^+$  and  $\text{Hg}_2^{2+}$               (C)  $\text{Pb}^{2+}$  and  $\text{Ag}^+$               (D) None of these

## Comprehension



6. What is (Z) ?  
 (A) It is salt of  $\text{Fe}^{3+}$  (B) It is a salt of  $\text{Sn}^{4+}$  (C) It is a salt of  $\text{Mn}^{2+}$  (D) It is a salt of  $\text{Cr}^{6+}$
7. What happens when X is kept in acidic medium ?  
 (A) Neutralisation reaction (B) Disproportionation reaction  
 (C) Double decomposition reaction (D) Pyrolytic reaction
8. **Statement-1** : Adding KCN to  $\text{CuSO}_4$  produces a white ppt which dissolves in excess KCN.  
**Statement-2** : With excess KCN,  $\text{CuSO}_4$  forms  $\text{K}_2[\text{Cu}(\text{CN})_4]$  complex.  
 (A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for statement-1  
 (B) Statement-1 is true, Statement-2 is true; Statement-2 is NOT a correct explanation for statement-1.  
 (C) Statement-1 is true, Statement-2 is false.  
 (D) Statement-1 is false, Statement-2 is true
9. **Statement-1** : Aluminium and zinc metal evolve  $\text{H}_2$  gas from NaOH solution.  
**Statement-2** : Several non-metals such as P, S, Cl, etc. yield a hydride instead of  $\text{H}_2$  gas from NaOH.  
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.  
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.  
 (C) Statement-1 is True, Statement-2 is False.  
 (D) Statement-1 is False, Statement-2 is True.

10.

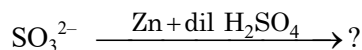
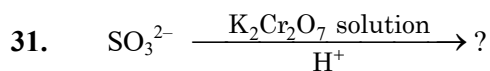
|     | Column-I (Reaction)                                                       |     | Column-II (Product)                    |
|-----|---------------------------------------------------------------------------|-----|----------------------------------------|
| (1) | $\text{Borax} \xrightarrow{\Delta}$                                       | (p) | BN                                     |
| (2) | $\text{B}_2\text{H}_2 + \text{H}_2\text{O} \longrightarrow$               | (q) | $\text{B}_2\text{H}_6$                 |
| (3) | $\text{B}_2\text{H}_6 + \text{NH}_3 (\text{excess}) \xrightarrow{\Delta}$ | (r) | $\text{H}_3\text{BO}_3$                |
| (4) | $\text{BCl}_3 + \text{LiAlH}_4 \longrightarrow$                           | (s) | $\text{NaBO}_2 + \text{B}_2\text{O}_3$ |

- (A) 1-r, 2-s, 3-p, 4-q (B) 1-s, 2-r, 3-p, 4-q (C) 1-s, 2-r, 3-q, 4-p (D) 1-q, 2-s, 3-r, 4-q

11. A white crystalline substance dissolves in water. On passing  $\text{H}_2\text{S}$  gas in this solution, a black precipitate is obtained. The black precipitate dissolves completely in hot  $\text{HNO}_3$ . On adding a few drops of conc.  $\text{H}_2\text{SO}_4$ , a white precipitate is obtained. This precipitate is not of :
- (A)  $\text{BaSO}_4$  (B)  $\text{SrSO}_4$  (C)  $\text{PbSO}_4$  (D)  $\text{CdSO}_4$
12. Colour of acidified  $\text{K}_2\text{Cr}_2\text{O}_7$  is not changed by :
- (A)  $\text{H}_2\text{O}_2$  (B)  $\text{Sn}^{2+}(\text{aq})$  (C)  $\text{HF}$  (D)  $\text{HBr}$
13. Which of the following will give positive chromyl chloride test :
- (A) Copper chloride,  $\text{CuCl}_2$  (B) Mercuric chloride,  $\text{HgCl}_2$   
 (C) Zinc chloride,  $\text{ZnCl}_2$  (D) Aniline hydro chloride,  $\text{C}_6\text{H}_5\text{NH}_2 \cdot \text{HCl}$
14. Suppose  $\text{Cu}^{2+}$  and  $\text{Pb}^{2+}$  both are present in a mixture, then they can be separated by :
- (A) Adding  $\text{KI}$  solution (B) Adding  $\text{NH}_3$  solution  
 (C) Adding  $\text{NaOH}$  solution (D) Adding dilute  $\text{HCl}$  (2M)
15. In electrolysis of  $\text{Al}_2\text{O}_3$  by Hall-Heroult process :
- (A) Cryolite  $\text{Na}_3[\text{AlF}_6]$  lowers the melting point of  $\text{Al}_2\text{O}_3$  and increases its electrical conductivity.  
 (B)  $\text{Al}$  is obtained at cathode and probably  $\text{CO}_2$  at anode  
 (C) Electrolysis is carried out in aqueous medium  
 (D) Anode consist of graphite
16. Heating which of the following salts in a dry test tube may cause a change in their colour ?
- (A)  $\text{ZnCO}_3$  (white) (B)  $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$  (red)  
 (C)  $\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$  (green) (D)  $\text{MnSO}_4$  (faint pink)
17. Iodine is formed when potassium iodide solution reacts with :
- (A)  $\text{ZnSO}_4$  (B)  $\text{CuSO}_4$  (C)  $\text{Cl}_2$  (D)  $\text{Br}_2$
18. Complexes formed in the MAC Arthur forest cyanide process are :
- (A)  $[\text{Au}(\text{CN})_2]^-$  (B)  $[\text{Ag}(\text{CN})_2]^-$  (C)  $[\text{Cu}(\text{CN})_4]^{2-}$  (D)  $[\text{Zn}(\text{CN})_4]^{2-}$
19.  $\Delta G^\circ$  vs  $T$  plot in the Ellingham diagram slopes upward for the reaction ?
- (A)  $\text{Mg}_{(\text{s})} + \text{O}_2 \rightarrow \text{MgO}_{(\text{s})}$  (B)  $2\text{Ag}_{(\text{s})} + \text{O}_2 \rightarrow \text{Ag}_2\text{O}_{(\text{s})}$   
 (C)  $\text{C}_{(\text{s})} + \text{O}_2 \rightarrow \text{CO}$  (D)  $\text{CO} + \text{O}_2 \rightarrow \text{CO}_2$
20. Which of the following oxyacid can not be obtained as a hydrolysed product of corresponding interhalogen compound, which have same oxidation number of underlined atom.
- (A)  $\text{HO}\underline{\text{F}}$  (B)  $\text{H}\underline{\text{I}}\text{O}_3$  (C)  $\text{H}\underline{\text{C}}\text{I}\text{O}_4$  (D)  $\text{H}\underline{\text{I}}\text{O}_4$

21. One gas bleaches the colour of flowers by reduction, while the other by oxidation, the two gases may be :-  
 (A)  $\text{Cl}_2$  (B)  $\text{H}_2\text{S}$  (C)  $\text{SO}_3$  (D)  $\text{SO}_2$
22. Select **CORRECT** for  $\text{HNO}_3$  reactions :-  
 (A) Cu produces  $\text{Cu}^{+2}$  and NO with dil.  $\text{HNO}_3$   
 (B) Ag produces  $\text{Ag}^+$  and  $\text{NO}_2$  with conc.  $\text{HNO}_3$   
 (C)  $\text{I}_2$  produces  $\text{HIO}_3$  and  $\text{NO}_2$  with conc.  $\text{HNO}_3$   
 (D) Zn produces  $\text{Zn}^{+2}$  and  $\text{NH}_4\text{NO}_3$  with very dil.  $\text{HNO}_3$
23. Select the correct statement(s) regarding reaction of  $\text{SO}_2$  with  $\text{PCl}_5$ .  
 (A) It is a redox reaction.  
 (B) One of the product is sulphuryl chloride.  
 (C) Both the products on addition of water produce strongly acidic solutions.  
 (D) Both the products have same hybridisation of central atom.
24. Which of the nitrates on strong heating do not leave metal as the residue?  
 (A)  $\text{AgNO}_3$  (B)  $\text{Pb}(\text{NO}_3)_2$  (C)  $\text{Cu}(\text{NO}_3)_2$  (D)  $\text{Al}(\text{NO}_3)_3$
25. Which of the following is/are not produced on reaction of  $\text{V}_2\text{O}_5$  with alkalis:  
 (A)  $\text{VO}_4^{3-}$  (B)  $\text{VO}_4^+$  (C)  $\text{VO}_2^+$  (D)  $\text{VO}_2^{2+}$
26. Select correct statements :-  
 (A) C–C bond length is longest in diamond among all crystalline allotropes of carbon.  
 (B)  $\text{BCl}_3 > \text{AlCl}_3 > \text{GaCl}_3 > \text{InCl}_3$ : Lewis acidic strength  
 (C) Potassium Ferrocyanide on reaction with concentration  $\text{H}_2\text{SO}_4$  gives a gas which burns with a blue flame  
 (D)  $\text{F}_2(\text{g})$  is used in etching of glass
27. Which of the following will undergo redox reaction on heating ?  
 (A)  $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$  (B)  $\text{Mg}(\text{NH}_4)\text{PO}_4$  (C)  $(\text{NH}_4)_2\text{SO}_4$  (D)  $\text{NH}_4\text{IO}_3$
28. Aqueous solution of boric acid is treated with salicylic acid. Which of the following statements is/are **incorrect** for the product formed in the above reaction ?  
 (A) no product will be formed because both are acids.  
 (B) product is 4-coordinated complex and optically resolvable.  
 (C) product is 4-coordinated complex and optically non-resolvable.  
 (D) there are two newly formed five membered rings.
29.  $\text{Na}_2\text{S}_2\text{O}_3$  is oxidised to  $\text{Na}_2\text{SO}_4$  by :  
 (A)  $\text{Cl}_2$  (B)  $\text{Br}_2$  (C)  $\text{I}_2$  (D)  $\text{KMnO}_4$

30. Which of the following statements is/are true about Mohr's salt ?
- (A) It decolourises  $\text{KMnO}_4$  (B) It gives  $\text{NH}_3$  on heating with  $\text{KOH}$ .  
 (C) It is a double salt (D) Oxidation state of iron is +3 in this salt



What is the difference between oxidation number of sulphur in product of both reactions ?

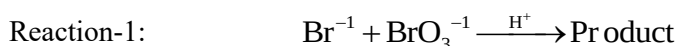
32. Amongst the following, the total number of compounds whose aqueous solution gives white precipitate with  $\text{Pb}^{2+}$  ions is :

$\text{KI}$ ,  $\text{NH}_3$ ,  $\text{Na}_2\text{CO}_3$ ,  $\text{K}_2\text{CrO}_4$ ,  $\text{NaCl}$ ,  $\text{Na}_2\text{SO}_3$ ,  $\text{Na}_2\text{S}$ ,  $\text{KNO}_3$ ,  $\text{NaClO}_4$

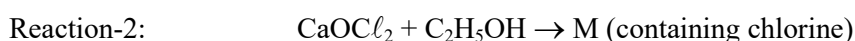
33. Find the number of reactions which produce  $\text{CO}_2$  as volatile product :-

- (1)  $\text{Na}_2\text{CO}_3 + \text{dil. H}_2\text{SO}_4$  (2)  $\text{KHCO}_3 + \text{dil. H}_2\text{SO}_4$   
 (3) Calcination of cerussite (4) Reaction of  $\text{H}_2\text{C}_2\text{O}_4$  with conc.  $\text{H}_2\text{SO}_4$   
 (5)  $\text{Fe}_2\text{O}_3 + \text{CO} \xrightarrow{500^\circ\text{C}, \Delta}$  (6)  $\text{ZnCO}_3 + \text{dil. HNO}_3$

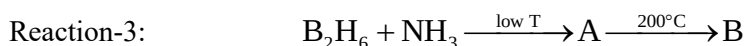
34. Consider following statements/reactions :



Let the oxidation state of bromine in product = x



Let the number of chlorine atom in a molecule of M = y

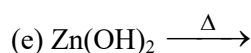
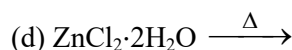
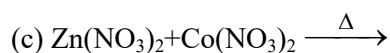
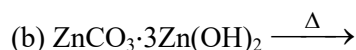
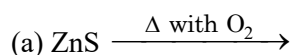


Let the number of  $\text{sp}^2$  hybridised orbitals in a molecule of B = z

Statement-1: In mineral Beryl,  $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$ , let number of oxygen atoms shared per tetrahedra = m

Find value of  $\frac{(x + y + z + m + 1)}{8}$  to nearest integer :

35. How many of the following reactions produces white residue (colour is observed after cooling).

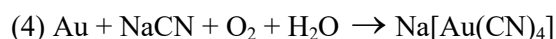


36. Number of incorrect statements is/are

(1) Froth floatation process for concentration of ores is an illustration of application of absorption.

(2) A sulphide ore is generally roasted to the oxide before reduction, because  $\text{CO}_2$  is thermodynamically less stable than  $\text{CS}_2$ .

(3) Addition of high proportions of manganese makes steel useful in making rails, because manganese imparts flexibility and softness.



(5) Magnesium is precipitated as  $\text{Mg(OH)}_2$  from sea water by treatment with lime and then dissolved in dilute  $\text{HCl}$  and reduced electrolytically in aqueous medium to get magnesium metal.

(6) During the extraction of aluminium by Hall-Heroult process, mixture of  $\text{Al}_2\text{O}_3$ ,  $\text{CaF}_2$  and  $\text{Na}_3\text{AlF}_6$  is used as electrolyte.

37. How many moles of a colourless and odourless gas (which burns with blue flame) are produced when one mole of  $\text{K}_4[\text{Fe(CN)}_6]$  is treated with excess of hot and concentrated sulphuric acid?

38. In how many of the following reactions  $\text{SO}_2$  gas is produced :-

(1) Heating of  $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

(2) Roasting of Fool's gold

(3) Heating  $\text{Cu}_2\text{O}$  with chalcocite

(4) Reaction of  $\text{Cu}$  with dil.  $\text{H}_2\text{SO}_4$

(5) Reaction of  $\text{Cu}$  with conc.  $\text{H}_2\text{SO}_4$

(6) Treatment of washing soda with dil.  $\text{H}_2\text{SO}_4$

(7) Hypo solution reacts with dil.  $\text{HCl}$

(8) White phosphorus react with  $\text{SO}_2\text{Cl}_2$

39. How many of the following anhydrous chlorides can't be obtained simply by heating their hydrated chlorides.  
MgCl<sub>2</sub>, CaCl<sub>2</sub>, AlCl<sub>3</sub>, ZnCl<sub>2</sub>, FeCl<sub>3</sub>
40. How many of the following will liberate H<sub>2</sub> gas on reaction with aq. NaOH solution (dilute or concentrated)  
NaH, Be, B<sub>2</sub>H<sub>6</sub>, Mg, F<sub>2</sub>, Cl<sub>2</sub>, P<sub>4</sub>(white), Si, Zn, BeH<sub>2</sub>
41. Among the following, the number of compounds that would require electrolysis process to get their respective metal from their metal compound is \_\_\_\_\_.  
NaCl, Cr<sub>2</sub>O<sub>3</sub>, MgCl<sub>2</sub>, Al<sub>2</sub>O<sub>3</sub>, CaCl<sub>2</sub>, Fe<sub>2</sub>O<sub>3</sub>
42. For the conversion of ore into their oxide, roasting process is used for how many given ores ?  
Sphalerite , Chalcocite , Fluorspar , Calamine , Azurite , Anglesite , Limonite , Cinnabar , Galena , Pyrolusite , Fool's gold
43. The number of compounds/elements oxidised by XeF<sub>2</sub> among following is:  
HF, HBr, HCl, HI, NH<sub>3</sub>, CrF<sub>2</sub>, Pt, S<sub>8</sub>
44. How many of the following acids may undergo disproportionation reaction on heating ?  
H<sub>2</sub>C<sub>2</sub>O<sub>4</sub>, H<sub>3</sub>PO<sub>2</sub>, H<sub>3</sub>PO<sub>3</sub>, HClO, HNO<sub>2</sub>, H<sub>2</sub>SO<sub>3</sub>, H<sub>2</sub>SO<sub>4</sub>, HClO<sub>3</sub>
45. SOCl<sub>2</sub> can react with how many of the following species to liberate SO<sub>2</sub>?  
H<sub>2</sub>O, HCl, C<sub>2</sub>H<sub>5</sub>OH, HBr, CH<sub>3</sub>COOH, HCN, H<sub>2</sub>SO<sub>4</sub>, H<sub>3</sub>PO<sub>4</sub>, D<sub>2</sub>O, HI, HF
46.  $\text{NaOH} + \text{PbO} \xrightarrow{\Delta} \text{X} + \text{H}_2\text{O}$   
 $\text{NaOH} + \text{SnO}_2 \xrightarrow{\Delta} \text{Y} + \text{H}_2\text{O}$   
 $\text{NaOH} + \text{H}_2\text{O} + \text{Al} \xrightarrow{\Delta} \text{Z} + \text{H}_2$   
Sum of the number of atoms present in one molecule each of X, Y, Z is .....
47. Amongst the following the total number of ions which produce colour in the solutions (in water) is.  
 ${}_{22}\text{Ti}^{3+}$ ,  ${}_{23}\text{V}^{4+}$ ,  ${}_{24}\text{Cr}^{3+}$ ,  ${}_{82}\text{Pb}^{2+}$ ,  ${}_{26}\text{Fe}^{2+}$ ,  ${}_{30}\text{Zn}^{2+}$ ,  ${}_{28}\text{Ni}^{2+}$ ,  ${}_{21}\text{Sc}^{3+}$ ,  ${}_{80}\text{Hg}^{2+}$
48. Which of the following will produce an insoluble precipitate with NH<sub>3</sub>(aq.)/H<sub>2</sub>S.  
BaCl<sub>2</sub>, NiSO<sub>4</sub>, Mohr's salt, AlCl<sub>3</sub>, Bi(NO<sub>3</sub>)<sub>3</sub>, Hg(CH<sub>3</sub>COO)<sub>2</sub>, AgNO<sub>3</sub>, SnCl<sub>2</sub>, CoSO<sub>4</sub>.

49. What is the oxidation number of Cr in the product of the reaction of  $K_2Cr_2O_7$  with  $H_2O_2/H^+$ , followed by boiling, followed by treatment with excess KOH.
50. How many of the following contain at least one iron atom in +2 oxidation state ?
- (a) Iron pyrites (b) Haematite  
(c) Magnetite (d) Brown ring complex  
(e) Sodium nitroprusside (f) Chromite  
(g) Ferrocene (h) Prussian blue  
(i) Wolframite

## ANSWER-KEY DPP # 06 (MAIN + ADVANCED)

- |                                                   |               |                                                                                        |               |             |
|---------------------------------------------------|---------------|----------------------------------------------------------------------------------------|---------------|-------------|
| 1. (B)                                            | 2. (D)        | 3. (C)                                                                                 | 4. (C)        | 5. (C)      |
| 6. (C)                                            | 7. (B)        | 8. (C)                                                                                 | 9. (C)        |             |
| 10. (1) → (s) ; (2) → (r) ; (3) → (p) ; (4) → (q) |               |                                                                                        | 11. (A,B,D)   | 12. (C)     |
| 13. (A,C,D)                                       | 14. (A,B,C,D) | 15. (A,B,D)                                                                            | 16. (A,B,C,D) | 17. (BCD)   |
| 18. (A,B,D)                                       | 19. (A,B,D)   | 20. (A,C)                                                                              | 21. (A,D)     |             |
| 22. (A,B,C,D)                                     | 23. (C,D)     | 24. (B,C,D)                                                                            | 25. (B,C,D)   | 26. (A,B,C) |
| 27. (A,D)                                         | 28. (A,C,D)   | 29. (A,B,D)                                                                            | 30. (A,B,C)   | 31. 8       |
| 32. 4                                             | 33. 6         | 34. 3                                                                                  | 35. 4         |             |
| 36. 5,[statements 1, 2,3, 4,5]                    |               | 37. (6) $[Fe(CN)_6]^{4-} + 12H^+ + 6H_2O \rightarrow Fe^{2+} + 6NH_4^+ + 6CO \uparrow$ |               |             |
| 38. 6                                             | 39. 4         | 40. 6                                                                                  | 41. 4         | 42. 5       |
| 43. 7                                             | 44. 6         | 45. 6                                                                                  | 46. 15        | 47. 5       |
| 48. 8                                             | 49. 3         | 50. 7                                                                                  |               |             |